Determinants of mortality in India: A quantile regression analysis

Ruchi Singh

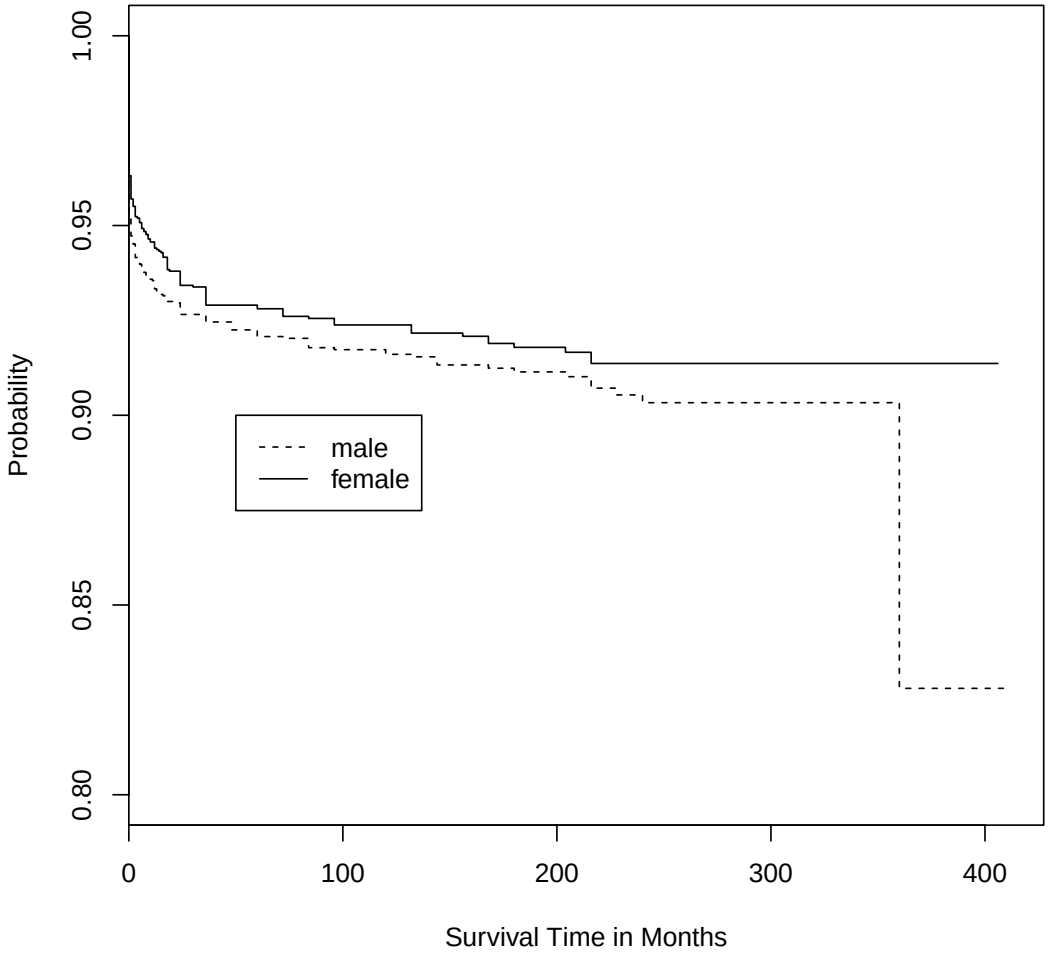
UIUC

May 6, 2014

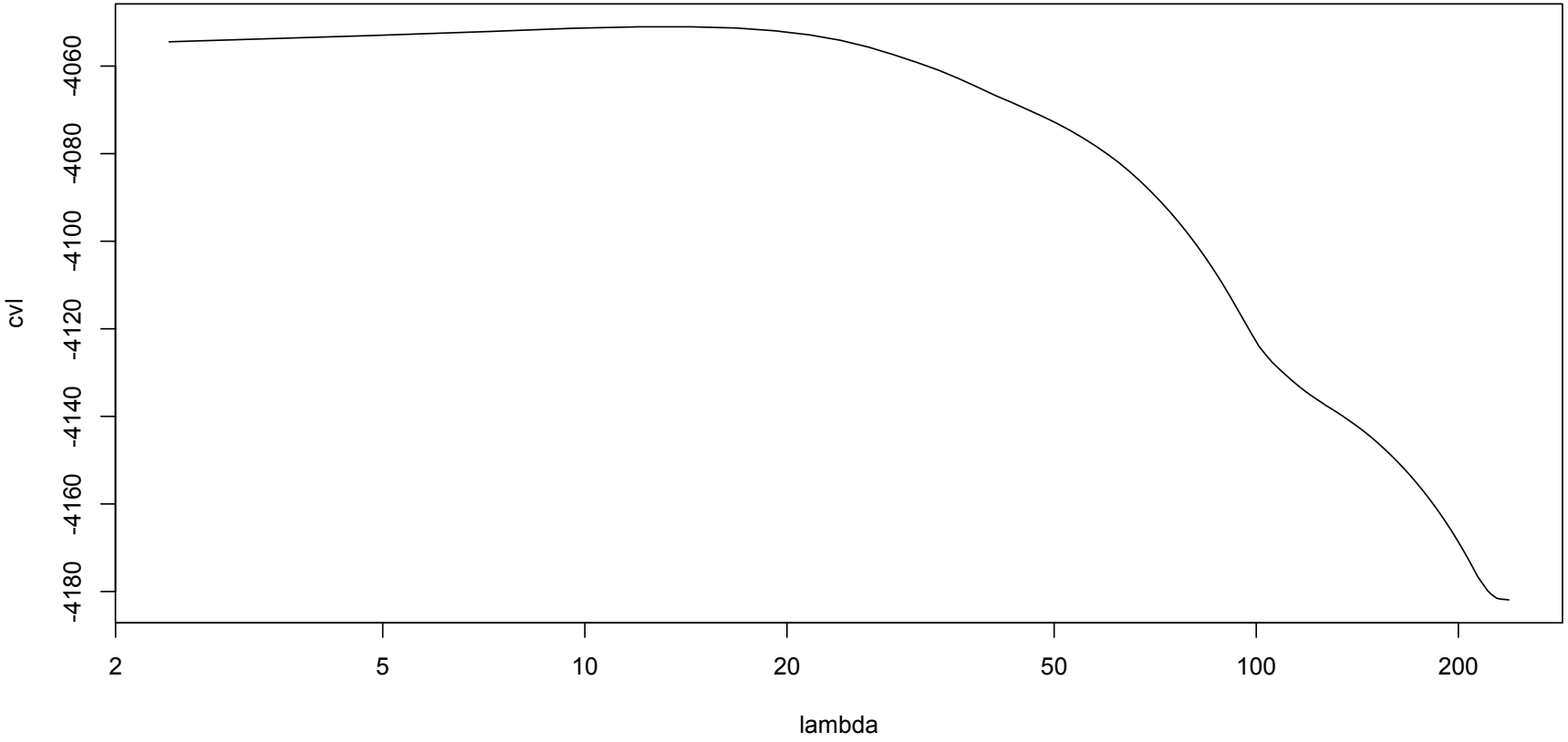
Determinants of mortality in India

- Child mortality is an indicator of country's socioeconomic development
- In 2012, 1.5 mn children under the age of 5 died in India which is an improvement compared to 2.5 mn in 2001
- However, India alone still accounts for 20% of child mortality worldwide
- India's under-5 child mortality rate in 2012 was 56 deaths per 1,000, far short of the Millennium Development Goal goal of reducing child mortality to 39 per 1,000 by 2015
- **Contribution of this paper:** This is the first paper to use LASSO and quantile regression to analyse child mortality in India
- **Data Source: Demographic Health Survey 2005-06**
  - Nationally representative sample of 124,385 ever married women in the 15-49 age group residing in the 33 states of India
  - For the mortality analysis I have a sample of 230,812 children in the 0-36 age group and a sample of 5517 for Andhra Pradesh

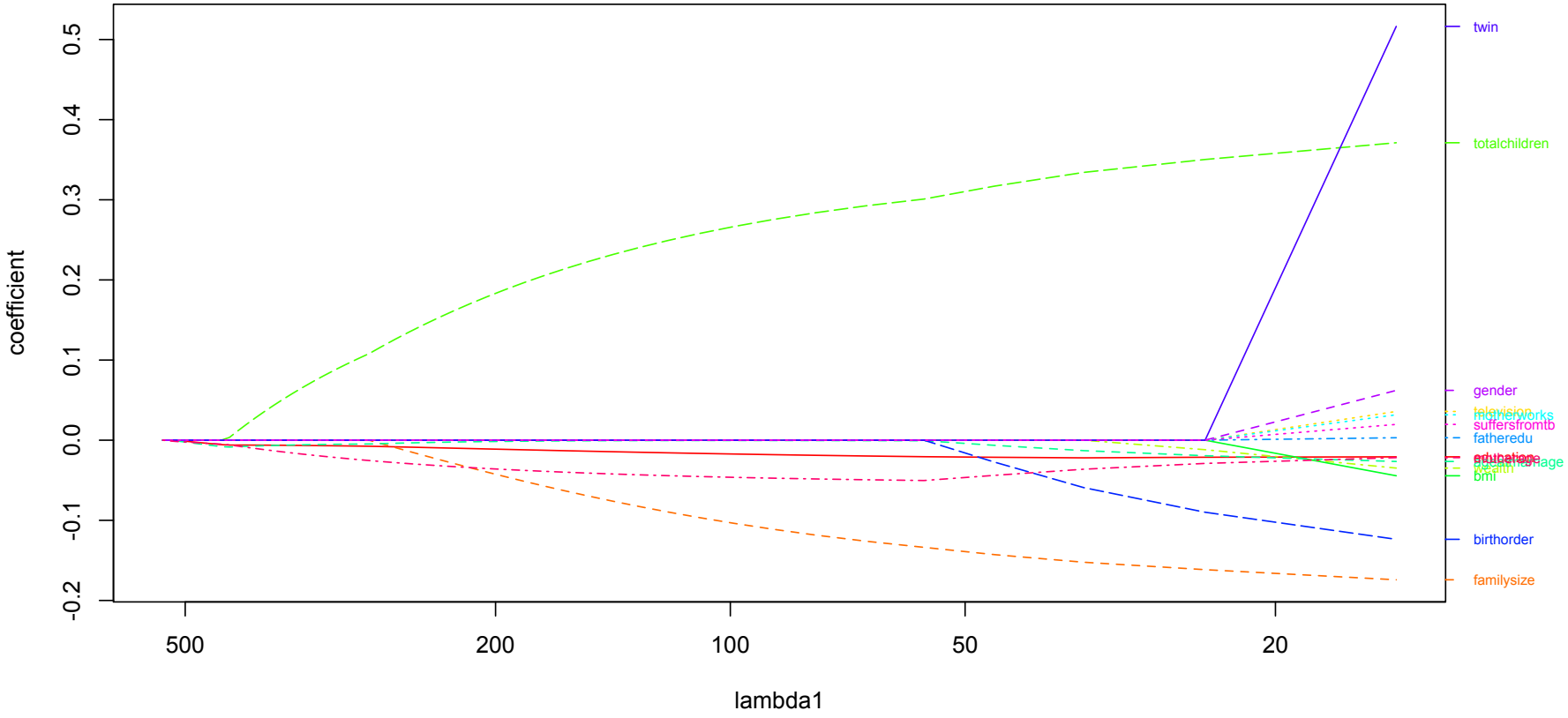
Kaplan-Meier estimates for Andhra Pradesh



Cross-Validated Likelihood



Effect of changing the tuning parameter on coefficients

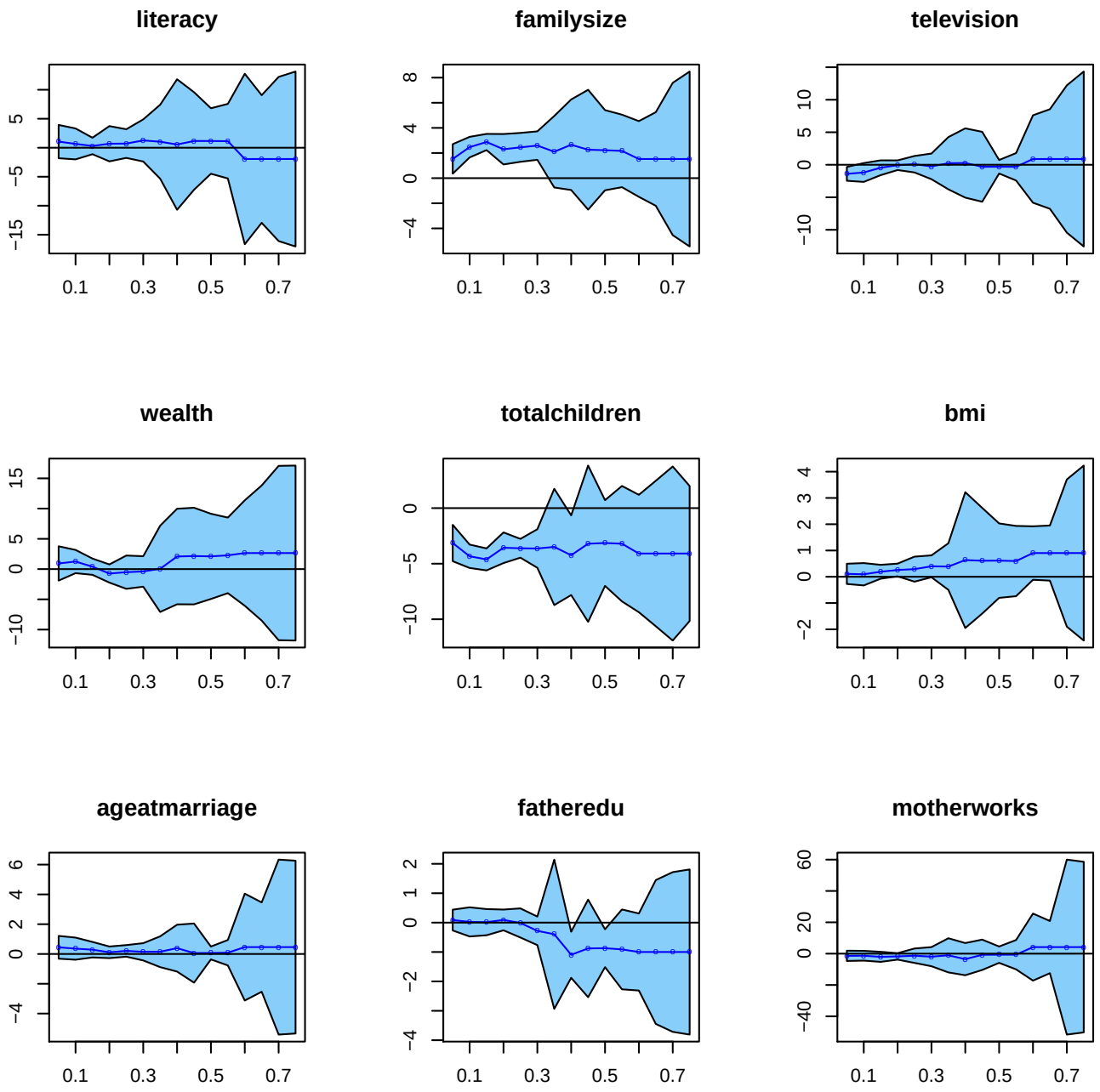


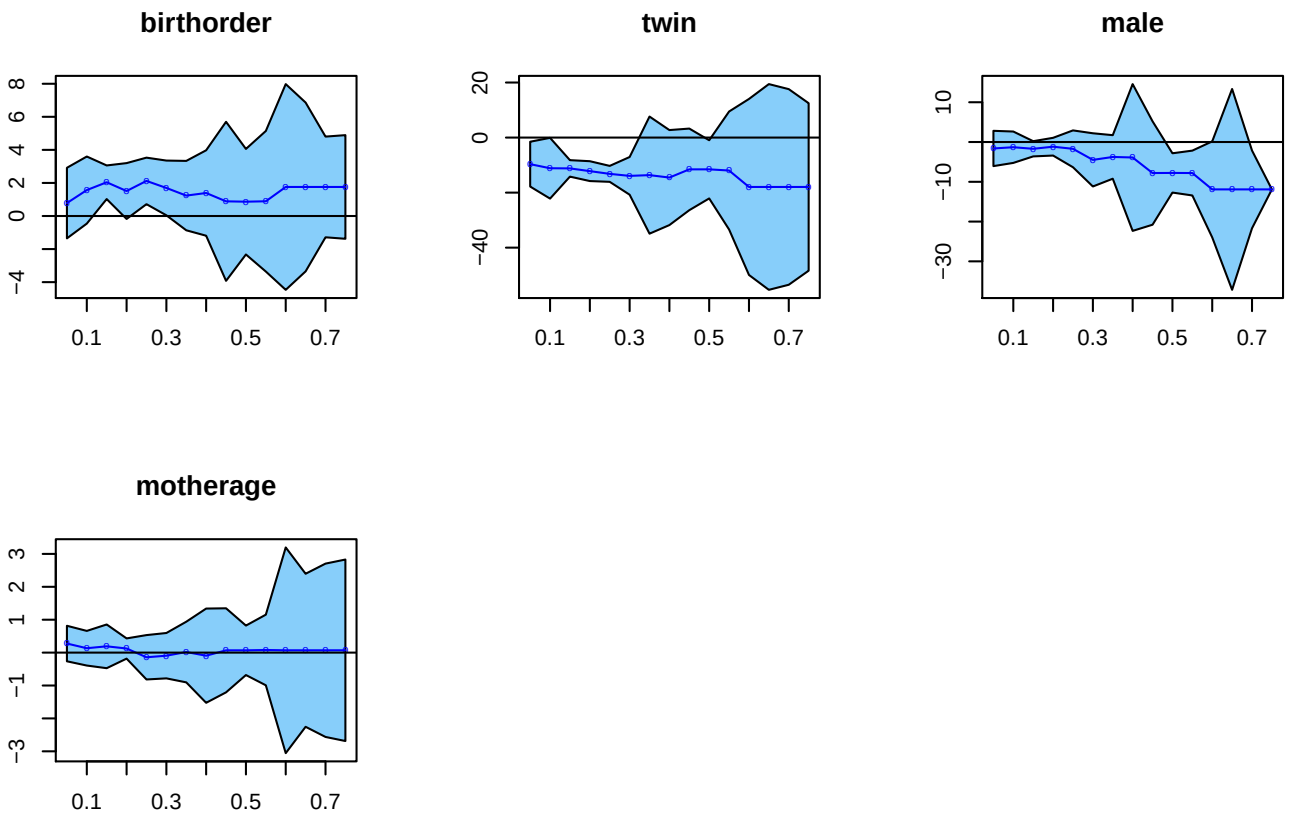
Cox Model: Andhra Pradesh

	coef	exp(coef)	se(coef)	z	p
literacy	-0.09	0.91	0.07	-1.30	0.19
familysize	-0.20	0.82	0.03	-7.76	0.00
television	0.08	1.09	0.04	1.98	0.05
wealth	-0.10	0.91	0.06	-1.71	0.09
totalchildren	0.41	1.51	0.03	13.47	0.00
bmi	-0.02	0.98	0.01	-1.89	0.06
ageatmarriage	-0.04	0.97	0.02	-1.79	0.07
fatheredu	0.01	1.01	0.01	0.84	0.40
motherworks	0.12	1.13	0.10	1.23	0.22
birthorder	-0.17	0.84	0.05	-3.56	0.00
twin	1.58	4.86	0.22	7.32	0.00
male	0.20	1.22	0.10	2.11	0.03
motherage	-0.02	0.99	0.02	-0.91	0.36

Censored Quantile Regression: Andhra Pradesh

Variable	tau= 0.2	tau= 0.4	tau= 0.6
literacy	0.687	0.552	-1.946
familysize	2.299	2.657	1.523
television	-0.068	0.272	0.886
wealth	-0.733	2.078	2.651
totalchildren	-3.565	-4.234	-4.085
bmi	0.259	0.635	0.903
ageatmarriage	0.111	0.391	0.462
fatheredu	0.091	-1.094	-0.999
motherworks	-1.754	-3.504	4.178
birthorder	1.510	1.391	1.757
twin	-12.197	-14.537	-17.948
male	-1.175	-3.890	-11.913
motherage	0.124	-0.094	0.070


literacy
5
−5
−15
0.1
0.3
0.5
0.7
familysize
8
4
0
−4
0.1
0.3
0.5
0.7
television
10
5
0
−10
0.1
0.3
0.5
0.7
wealth
15
5
0
−10
0.1
0.3
0.5
0.7
totalchildren
0
−5
−10
0.1
0.3
0.5
0.7
bmi
4
3
2
1
0
−2
0.1
0.3
0.5
0.7
ageatmarriage
6
4
2
0
−4
0.1
0.3
0.5
0.7
fatheredu
2
1
0
−2
−4
0.1
0.3
0.5
0.7
motherworks
60
20
0
−40
0.1
0.3
0.5
0.7


birthorder
8
6
4
2
0
−4
0.1
0.3
0.5
0.7
twin
20
0
−40
0.1
0.3
0.5
0.7
male
10
−10
−30
0.1
0.3
0.5
0.7
motherage
3
2
1
−1
−3
0.1
0.3
0.5
0.7

Under 5 mortality: For the entire country

Table : Cox Proportional Hazard Model

Variable	coef	exp(coef)	se(coef)	z	p
state	-0.003	0.997	0.000	-3.865	0.000
religion	-0.107	0.898	0.008	-12.809	0.000
motheredu	-0.032	0.968	0.002	-13.674	0.000
familysize	-0.143	0.866	0.003	-43.563	0.000
television	0.017	1.018	0.016	1.067	0.290
wealth	-0.091	0.912	0.007	-12.122	0.000
totalchildren	0.290	1.338	0.003	86.771	0.000
bmi	-0.016	0.983	0.001	-8.484	0.000
ageatmarriage	0.039	1.040	0.002	14.762	0.000
motherworks	0.028	1.029	0.014	1.977	0.048
twin	1.509	4.522	0.031	48.421	0.000
male	0.118	1.126	0.013	8.623	0.000
motherage	-0.075	0.927	0.001	-47.415	0.000

Under 5 mortality: For the entire country

Table : Censored Quantile Regression

Variable	tau= 0.2	tau= 0.4
state	0.018	0.028
religion	0.486	0.527
motheredu	0.239	0.544
familysize	1.110	0.651
television	-0.068	-0.065
wealth	0.385	0.653
totalchildren	-1.566	-0.932
bmi	0.102	0.087
ageatmarriage	-0.232	-0.121
fatherworks	-0.772	0.675
motherworks	-0.003	0.142
twin	-14.984	-5.532
male	-0.545	-0.355
motherage	0.387	0.254